

Mohammed Dasouqi

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Transferable Iqama — available to join without new visa sponsorship

PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on, end-to-end experience across the machine learning lifecycle — from data preprocessing and model training to evaluation and deployment. Built deep learning systems for medical image analysis (PyTorch, YOLO, EfficientUNet++), NLP and semantic-search pipelines (Sentence-BERT, LlamaIndex, local LLM deployment), and data-driven backend applications (FastAPI, SQL). Strong foundation in data structures, algorithms, object-oriented software design, and software engineering principles. Seeking an AI/ML Engineer or Data Engineer role building scalable, production-grade intelligent systems.

EDUCATION

University of Nottingham

Bachelor of Science (Honours), Software Engineering
September 2021 – July 2025

- Classification: Upper Second Class Honours (2:1)

TECHNICAL SKILLS

- **Machine Learning & AI:** PyTorch, YOLO (object detection), EfficientUNet++ (image segmentation), Sentence-BERT, NLP question-answering models, LlamaIndex, Ollama (local LLM deployment), semantic embeddings, model evaluation (precision, recall, F1-score, Dice, IoU)
- **Programming Languages:** Python, Kotlin, C++, PHP, JavaScript, SQL, HTML/CSS
- **Backend & Data:** FastAPI, SQLAlchemy, Retrofit, REST API design, asyncio/aiohttp, MySQL, Firebase/Firestore
- **Core Foundations:** Data structures & algorithms, object-oriented design, software development lifecycle (SDLC), design patterns (Observer, Repository, MVVM)
- **Tools & Platforms:** Git/GitHub, TensorBoard, CUDA, VS Code, JetBrains IDEs, Google Maps/Places/Directions APIs

PROJECTS

Brain Tumor Detection & Segmentation — Final Year Project | *PyTorch 2.6, YOLO, RepVGG-GELAN, EfficientUNet++, EfficientNet-B0, CUDA*

- Designed and trained a two-stage deep learning pipeline on the BR35H MRI dataset (800 scans) combining a custom RepVGG-GELAN detector with a nested EfficientUNet++ segmentation network; segmentation model achieved a Dice score of 93.75%, IoU of 89.44%, precision of 95.09%, and recall of 95.92% on the held-out test set
- Detection model met or exceeded the results of the original RepVGG-GELAN paper (Precision 98.6% vs 98.2%, AP50:95 73.0% vs 72.3%), trained entirely from scratch without pretrained detection weights
- Implemented annotation quality filtering — excluding imprecise circular and elliptical ground truth masks from training — which improved segmentation Dice score by 1.5% and eliminated a systematic 'circular effect' bias in predicted masks
- Built the full ML engineering pipeline from scratch: dataset conversion scripts (YOLO format), custom detection-to-segmentation cropping bridge, joint data augmentation, TensorBoard logging, and per-epoch visual validation of predicted vs. ground truth masks

Intelligent Web Crawler (AI Agent) | *Python, asyncio/aiohttp, BeautifulSoup, Sentence-BERT, NLP*

- Built a budget-aware asynchronous web crawler combining Sentence-BERT semantic scoring, anchor-text relevance weighting, and a BERT-based QA snippet extraction pipeline to dynamically prioritize URLs, replacing static breadth-first traversal strategies
- Implemented a closed-loop online fine-tuning system that periodically re-trained both the embedding and snippet-extraction models on collected data mid-crawl, improving relevance scores in 80% of experimental runs
- Benchmarked across 25 runs, achieving average precision of 95.4%, recall of 85.6%, and F1-score of 90.1%, outperforming breadth-first and TF-IDF baseline crawlers while maintaining semantic diversity above 60% throughout all runs

Chat SQL Agent (AI-Powered Web App) | *Python, FastAPI, LlamaIndex, Ollama, SQLAlchemy*

- Built a natural-language-to-SQL application that lets non-technical users query relational databases through a RESTful FastAPI backend, removing the need to write SQL manually
- Integrated LlamaIndex with a locally hosted Ollama LLM for schema reasoning and adaptive SQL generation, enabling fully offline, context-aware query translation
- Implemented semantic embeddings for local inference, removing dependency on external API calls and eliminating per-query API costs

AutoConnect (Android App) | *Kotlin, Jetpack Compose, Firebase Firestore, Retrofit, Google Maps/Directions/Places API*

- Designed and built a feature-complete roadside-assistance Android app with a 4-layer service-oriented architecture (Data, Service, UI, Utility), 8 screens, and role-based access control for customers, drivers, and admins
- Implemented real-time driver-to-customer location tracking via Firestore listeners (Observer pattern) with 30-second background location sync, plus push notifications for live tow-request status updates
- Integrated Retrofit-based REST clients for Google Directions, Places, and Maps SDKs within a Repository pattern (single source of truth), separating all external API interactions from the UI layer
- Built the full UI in Jetpack Compose (no XML) with reusable components (CustomAppBar, MapViewComponent, TechnicianGrid) and a ViewModel layer, enabling clean separation of state management from rendering logic

Personal Portfolio Website | *HTML, CSS, JavaScript (Vanilla — no frameworks)*

- Built a fully custom single-page portfolio from scratch using only vanilla HTML, CSS, and JavaScript — no frameworks, no build step, no dependencies — demonstrating deliberate low-level frontend engineering rather than reliance on abstraction layers
- Implemented a command palette (Ctrl/Cmd+K), animated neural-network background, scroll-reveal animations, and three switchable colour themes with localStorage persistence, all within a single data-driven config file
- Built an interactive MRI segmentation demo embedded directly in the portfolio, visualizing the Brain Tumor Detection FYP pipeline output for non-technical audiences

Online Booking System (Full-Stack Web App) | *PHP, MySQL, HTML/CSS*

- Built a full-stack booking platform with user authentication, payment processing, and role-based admin management
- Designed reusable backend and frontend components, reducing duplication and speeding up feature development
- Collaborated within a cross-functional team using iterative development to plan, build, and refine system features

EXPERIENCE

Automotive Diagnostics Technician — *Digital Engineering, Riyadh, Saudi Arabia* *May 2026 (1 month)*

- Diagnosed vehicle electronic and software faults using OBD-II diagnostic tools and manufacturer software, identifying root causes across multiple vehicle systems

- Performed ECU parameter calibration and software updates to bring vehicle systems back to manufacturer specification
- Applied a systematic, evidence-based troubleshooting methodology, directly transferable to debugging and root-cause analysis in software and ML systems

LANGUAGES

- Arabic — Native | English — Fluent (bilingual)